

TECHNICAL SKILLS

AI Agents & LLMs	LLMs, Multi-Agent Systems, Tool Calling (MCP), RAG, Prompt Grounding, Context & Memory Graphs, Fine-Tuning
Voice & Multimodal	LiveKit (WebRTC), Silero VAD, Streaming STT/TTS (Deepgram Nova-3, ElevenLabs), Turn Detection, TensorFlow, PyTorch
Software Engineering	Python (FastAPI, asyncio), TypeScript/JavaScript (Node.js, React, Vite), Kotlin (Android Compose), C/C++ (FreeRTOS), REST APIs
Hardware & Systems	3D CAD Modeling (STL), Wearable Packaging, ESP32-S3 Firmware (I2S DMA, Opus, OTA), Kubernetes Helm, Git/GitHub, Linux

EXPERIENCE

Founder — Aura	Jun 2025 – Present
<i>Wearable AI · Embedded Systems · Edge Integration · Open Source</i>	<i>Pune, India</i>
- Designed custom 3D CAD enclosures packaging Sseed ESP32-S3 Sense, camera, LiPo, and mic into an ergonomic pendant. - Wrote FreeRTOS event loops for I2S DMA audio capture at 16 kHz with Opus on-device compression and OTA firmware updates (<code>firmware/src/mic.cpp</code>). - Integrated physical device with open-source Omi ecosystem via FastAPI backend (memory, transcription, graph routers) and Deepgram ASR. - Diagnosed Electron WebGL GPU memory leak on Windows caused by CSS backdrop filters over 3D canvas (authored Issue #8438, merged downstream in PR #8902). - Authored offline agglomerative hierarchical clustering (SciPy cosine linkage, 52 unit tests) to solve speaker drift in 3+ person meetings (PR #8919, adopted downstream in PR #12471).	
AI & ML Intern — AICTE via Google	Oct 2024 – Dec 2024
<i>Computer Vision · TensorFlow · ML Kit · Mobile Edge Deployment</i>	<i>Pune, India</i>
- Built and evaluated lightweight on-device vision models with TensorFlow and Google ML Kit for mobile edge classification under memory constraints. - Applied transfer learning and data augmentation pipelines to optimize model inference latency and accuracy on mobile hardware.	
Freelance Developer — AI & Automation	Sep 2024 – Present
<i>Workflow Automation · LLM Tool Calling · Rapid Prototyping</i>	<i>Remote</i>
Built custom AI agent workflows and automation scripts for developer projects, integrating LLM tool calling and automated verification gates.	

PROJECTS

Mia — Personal AI Engineering OS	github.com/thesohamdatta/Mia
<i>TypeScript · Bun · Multi-Host LLM Adapters · Vitest · JSONL State Engine</i>	
- Built a compiled TypeScript CLI (single Bun binary) automating a 5-stage AI engineering harness (<code>grill</code> → <code>spec</code> → <code>plan</code> → <code>review</code> → <code>ship</code>) with unified adapters for Claude, Codex, and local models. - Implemented immutable JSONL state tracking, automated verification gates (lint, typecheck, Vitest), and eliminated background daemon architecture in favor of stateless execution (ADR-0001).	
Voice AI Systems — Real-Time Voice Agents & Mobile Clients	GitHub / Voice-AI
<i>LiveKit WebRTC · Silero VAD · ElevenLabs · Python · Kotlin (Android Jetpack Compose)</i>	
- Built real-time WebRTC conversational voice agents using LiveKit with client-side Silero VAD for instant barge-in interruption and semantic turn detection. - Developed John (native Android Kotlin/Compose client with LiveKit tool calling) and voicecoder (VS Code extension with multi-provider LLM fallback and real-time token tracking).	
LLM-Council — Multi-Model Consensus Engine	github.com/thesohamdatta/LLM-Council
<i>Python · FastAPI · React · Vite · Async Multi-Provider Orchestration</i>	
- Built a full-stack multi-LLM consensus engine querying OpenAI, Anthropic, Gemini, and Groq concurrently to cross-evaluate and synthesize peer-reviewed answers. - Implemented a 3-stage critique pipeline (Analyst → Skeptic → Synthesizer) with an interactive React/Vite frontend visualizing step-by-step model agreement.	

EDUCATION

University of Pune (SPPU)	Pune, India
<i>Bachelor of Engineering (B.E.) in Artificial Intelligence & Machine Learning</i>	<i>2022 – Jun 2026</i>
Core Coursework: Computer Vision, Edge AI Inference, LLMs & Multi-Agent Systems, Deep Learning, Distributed Systems, Algorithms	